

Which density belongs in the NPC optical depth?

Short answer: use the number density of the collision targets, not automatically the total mass density divided by the proton mass.

For matter made of free protons and neutrons:

$$n_b = \frac{\rho}{m_p}$$

$$n_p = Y_e n_b, \quad n_n = (1 - Y_e) n_b$$

Here n_b is the total baryon number density. The electron fraction Y_e sets the proton fraction only while the material is represented as free nucleons and charge neutrality applies.

The target depends on the projectile:

$$\tau_{n \rightarrow p} = n_p \sigma_{np} \frac{\Delta r}{\Gamma_a}$$

$$\tau_{p \rightarrow n} = n_n \sigma_{pn} \frac{\Delta r}{\Gamma_a}$$

Therefore one composition-independent optical depth cannot describe both legs of a neutron--proton converter cycle.

Why the distinction matters for BNS ejecta

Example: neutron-rich material with $Y_e = 0.10$

$$n_p = 0.10 n_b, \quad n_n = 0.90 n_b$$

$$\tau_{n \rightarrow p} = 0.10 \tau_{\text{baryon}}$$

If the present calculation gives $\tau_{\text{baryon}} = 4.4\text{--}6$, the free-proton approximation gives $\tau_{(n \rightarrow p)} = 0.44\text{--}0.60$ for $Y_e = 0.10$. This lies inside the $\tau_{pn} = 0.1\text{--}2$ range explored by Kashiya et al. (2013).

Important limitation

After nuclei form, $1-Y_e$ is NOT the free-neutron mass fraction. A realistic composition must distinguish free protons, free neutrons, and nuclei:

$$X_p^{\text{free}}, \quad X_n^{\text{free}}, \quad X_{\text{nuclei}}$$

Collisions with heavy nuclei need appropriate nuclear cross sections and fragmentation physics. Treating them as free pn collisions is not justified.

Recommended quantities for the Monte Carlo int

Export separately at every trajectory sample:

1. Mass density: ρ
2. Total baryon density: $n_b = \rho/m_p$
3. Electron fraction: Y_e
4. Free-proton density: $n_p = X_{p_free} n_b$
5. Free-neutron density: $n_n = X_{n_free} n_b$
6. $\tau_{(n \rightarrow p)}$ and $\tau_{(p \rightarrow n)}$
7. τ_{baryon} only as a reference upper-bound calculation

If only Y_e is available

Use $X_{p_free} = Y_e$ and $X_{n_free} = 1 - Y_e$ only as an explicitly labelled free-nucleon approximation. Give the Monte Carlo both species densities so it can select the target appropriate to the projectile at each step.

Do not mix two different optical depths

τ_{pn} : hadronuclear collisions

τ_γ : photon/electron breakout opacity

Changing the pn target fraction does not automatically change the shock-breakout opacity. The latter requires its own composition-dependent model.